

Proof by Contraposition: $\neg(Q(n) \vee R(n)) \rightarrow \neg P(n)$

or, by De Morgan's Law, $\neg Q(n) \wedge \neg R(n) \rightarrow \neg P(n)$

$\neg P(n): n \neq ab$ $\neg Q(n): a > \sqrt{n}$ $\neg R(n): b > \sqrt{n}$

Let, $\neg Q(n) \wedge \neg R(n)$ be True. By Defn. of AND, $\neg Q(n)$ and $\neg R(n)$ are each True respectively.

~~QED~~ $\therefore a > \sqrt{n}$ and $b > \sqrt{n} \therefore ab > \sqrt{n} \cdot \sqrt{n} \Rightarrow \boxed{ab > n.}$

$\therefore n \neq ab \quad \square$

Vacuous and Trivial Proofs: We can quickly prove that a conditional statement $P \rightarrow Q$ is true when we know that P is false, because we know that $P \rightarrow Q$ must be true when P is false. Consequently, if we can show that P is false, then we have a proof, called a Vacuous proof, of the conditional statement $P \rightarrow Q$.

Example 5: Show that the proposition $P(0)$ is true, where $P(n)$ is "If $n > 1$, then $n^2 > n$ " and the domain consists of all integers.

Soln: Note that $P(0)$ is "If $0 > 1$, then $0^2 > 0$." We can show using a vacuous proof. Indeed the hypothesis $0 > 1$ is false. This tells us that $P(0)$ is automatically true. $\square$

Remark: The fact that the conclusion of this conditional statement, $0^2 > 0$, is false is irrelevant to the truth value of the conditional statement, because a conditional statement with a false hypothesis is guaranteed to be true.

Example 6: Prove that if n is an integer with $10 < n < 15$ which is a perfect square, then n is also a perfect cube.